

Asia Emerging Robotics

Asia Emerging Robotics: The Barometer (June/July 2026)



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In this note, we update the key events in humanoid robot industry, market sentiment, and important data points in June & July 2026. Readers can download the complete barometer dataset here: [Asia Emerging Robotics Barometer](#).

Key events (Exhibit 1). In July, the most consequential event for the humanoid robotics industry was the Trump administration's ban on new Chinese humanoid robots ([link](#)), which we view as the beginning of US-China decoupling in robotics. Among Western humanoid players, Tesla indicated that Optimus Gen 3 will enter production soon, although the initial ramp-up is likely to be gradual. The company also suggested that Gen 4 would be more vertically integrated, while we think it is not easy to shed its Chinese partners, just as it still needs them for its vehicles. Meanwhile, Boston Dynamics showcased its Atlas robot at FIFA World Cup ([link](#)). In China, most leading Chinese humanoid players showcased their latest developments at the World AI Conference (WAIC). Industry focus is increasingly shifting from locomotion toward robotic "brain" and practical applications. Although commercialization remains at an early stage, Chinese players appear to have entered a new phase of trial-and-error characterized by rapid iteration and real-world validation.

Market Sentiment. Sentiment toward humanoid robots plunged in July amid a broader sell-off in technology names (Exhibit 2 to Exhibit 4). Negative robotics headlines (e.g., the US ban on new Chinese humanoid robots and Tesla's potential move toward greater vertical integration) further weighed on sentiment, pushing sector valuations back toward levels seen during the 2025 sentiment trough. In our view, the recent pullback offers an **attractive entry point** into humanoid-related stocks.

Monthly data (Exhibit 5 to Exhibit 18) that underpins the fundamentals:

- **Shuanghuan:** China xEV wholesales volume +22% YoY in June (+13% YoY in May); China robotic vacuum sales volume +23% YoY in June (+0.3% YoY in May); China industrial robot production volume +28% YoY in June (+28% YoY in May).
- **Hesai:** Shipment volume of Hesai's long-range ADAS LiDAR +82% YoY in June (+69% YoY in May); Hesai's market share was 45% in June (45% in May).
- **Tuopu:** China xEV wholesales volume +22% YoY in June (+13% YoY in May); Shipment volume of Tuopu's air suspension -5% YoY in June (+39% YoY in May); Aluminum price +11% YoY in July (+16% YoY in June).
- **Sanhua:** China xEV wholesales volume +22% YoY in June (+13% YoY in May); China domestic sales and export volume of air conditioners was -7% YoY in June (-11% YoY in May) and the volume was expected to -13% YoY in 3Q26 (based on production plan); Copper price +31% YoY in July (+32% YoY in June); Aluminum price +11% YoY in July (+16% YoY in June).
- **Leader Drive:** China industrial robot production volume +28% YoY in June (+28% YoY in May); China robotic reducer production volume +57% YoY in 1H2026 (+64% YoY in 2025).

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DETAILS

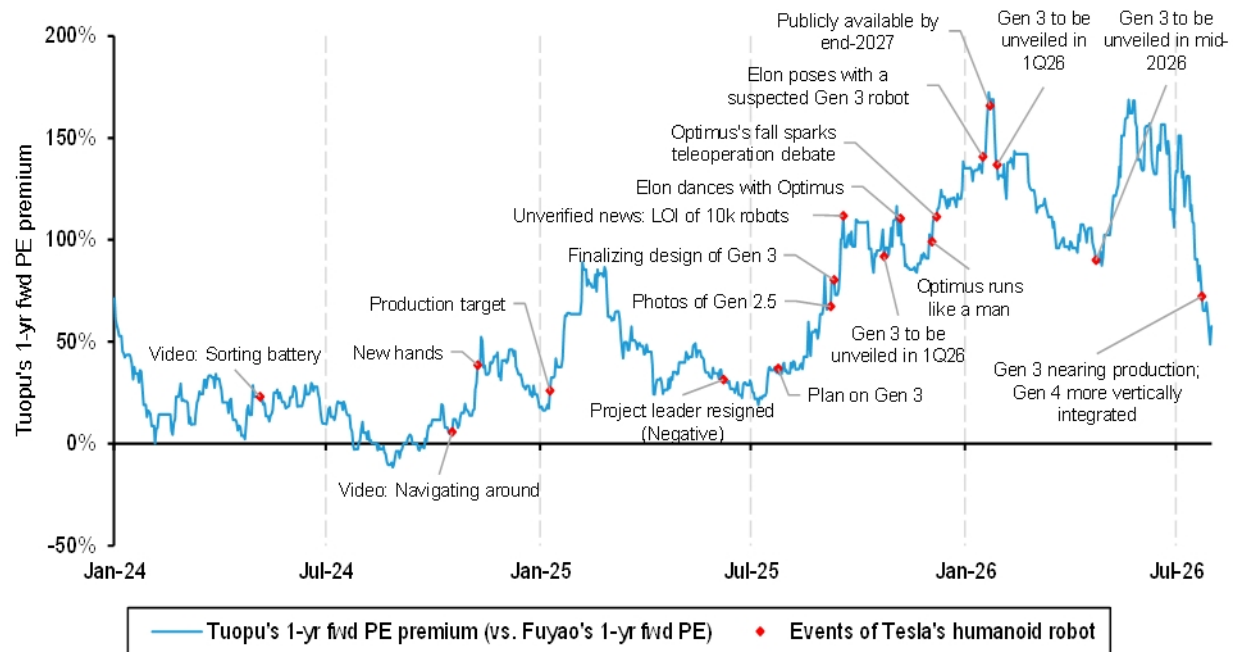
EXHIBIT 1: Key events in the humanoid robotics industry (Jun & Jul 2026)

Date	Region	Institute	Ticker	Events
Jun 1, 2026	China	Unitree	Unlisted	Unitree clears STAR market listing review in record 73 days.
Jun 2, 2026	China	UBTech	9880.HK	UBTECH launches pre-sales for full-size hyper-bionic humanoid robot.
Jun 3, 2026	China	BYD	1211.HK	BYD confirms humanoid robot development, says future sales could use dealer network.
Jun 3, 2026	USA	U.S. Congress	-	A group of bipartisan US lawmakers has introduced the GUARD Act to ban dangerous Chinese robots.
Jun 4, 2026	USA	U.S. Congress	-	U.S. Senators Dave McCormick (R-PA), John Hickenlooper (D-CO), Todd Young (R-IN), and Martin Heinrich (D-NM) today introduced the bipartisan National Commission on Robotics Act.
Jun 9, 2026	China	Chinese government	-	China's MIT and SASAC launch 2026 humanoid robot real-scene training initiative.
Jun 15, 2026	China	Seres	601127.CH	Chinese automaker unveils Xiaosai humanoid robot for inspection and production roles.
Jun 21, 2026	China	UBTech	9880.HK	UBTECH presales ~5,000 units of its full-size hyper-bionic humanoid robot.
Jun 22, 2026	USA	NVIDIA	NVDA.US	NVIDIA releases Halos, a full-stack safety system for robotics.
Jun 22, 2026	China	Chinese government	-	China plans to introduce standards to help humanoid robots enter more real-world applications (e.g. power substations and home appliance manufacturing).
Jun 24, 2026	China	Unitree	Unlisted	Unitree cuts the R1 humanoid price by 25%, from RMB39,900 (~\$5,900) to RMB29,900 (~\$4,400).
Jun 24, 2026	USA	Agility Robotics	Unlisted	Agility Robotics plans to go public through merger with Churchill Capital Corp XI.
Jun 25, 2026	USA	Figure	Unlisted	Figure 03 robots start new use cases at BMW Group's Plant Spartanburg, working on complex sequencing applications in logistics.
Jun 28, 2026	China	AgiBot	Unlisted	AgiBot announces they have mass produced 15,000 units of humanoid robots accumulatively.
Jun 29, 2026	China	AgiBot	Unlisted	AgiBot's robots achieve a 99.99% success rate in quality inspection and material-handling tasks during a six-day livestreamed factory demonstration.
Jun 30, 2026	China	UBTech	9880.HK	UBTECH launches hyper-bionic humanoid robot UWORLD U1 with claimed 13,361 orders.
Jul 1, 2026	USA	Tesla	TSLA.US	Elon Musk shares Tesla Optimus production team photo, while says Optimus production will be 'extremely slow' in the beginning.
Jul 2, 2026	China	Unitree	Unlisted	Unitree receives final approval for a listing on the Shanghai STAR Market.
Jul 6, 2026	USA	Boston Dynamics	Subsidiary of Hyundai Motor Group	Boston Dynamics brings its legged robots to the FIFA World Cup.
Jul 10, 2026	China	Unitree	Unlisted	Unitree G1 humanoid performs live animal surgery in study.
Jul 23, 2026	USA	Tesla	TSLA.US	Tesla 2Q earnings call: • Optimus production will start soon, but the ramp is likely to follow an S-curve. • The initial Optimus builds will be used in its Optimus Academy for training data collection and further functionality development. • Optimus V4 (the next generation) will be more vertically integrated.
Jul 28, 2026	USA	U.S. Federal Communications Commission	-	Trump administration bans new Chinese humanoid robots.
Jul 30, 2026	USA	Google	GOOGL.US	Google introduces Gemini Robotics 2 with 'Whole Body Intelligence'.

Source: Industry news, companies' websites, Bernstein analysis

EXHIBIT 2: **Tuopu: Valuation premium attributed to Tesla & humanoid robots**

Tuopu's valuation premium vs. Events of Tesla humanoid robot

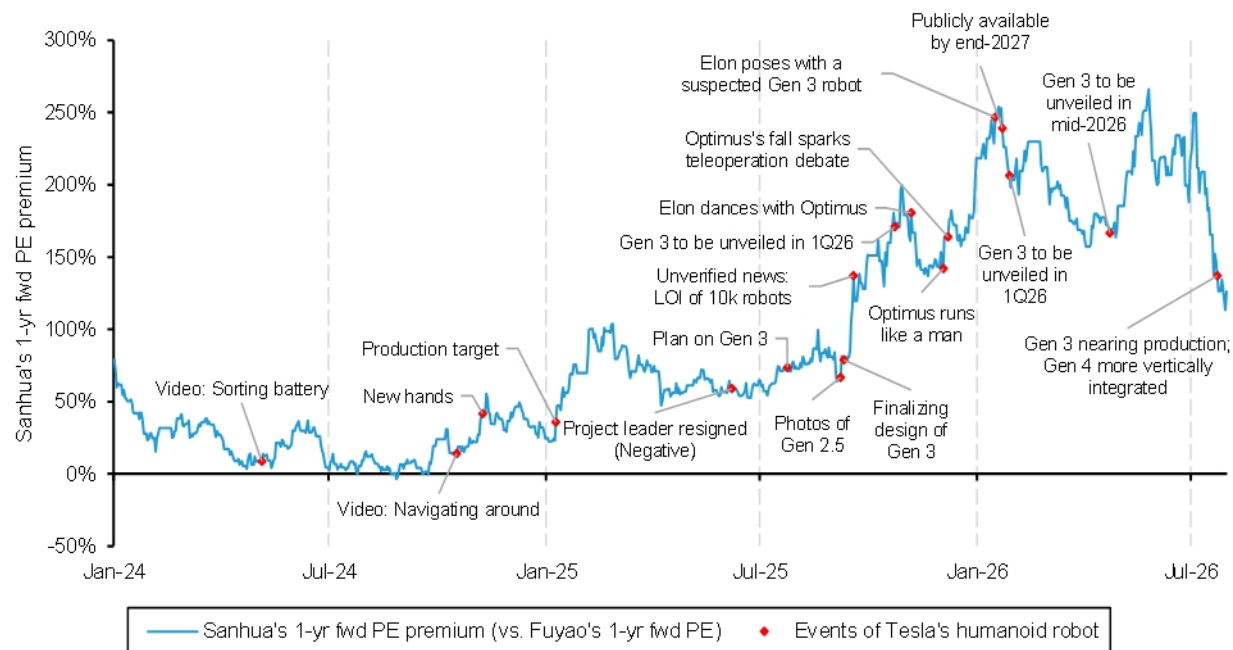


Note: We use Fuyao Glass (600660.CH/3606.HK) as the benchmark. Fuyao Glass is not covered by Bernstein.

Source: Tesla, industry news, Tuopu, Fuyao Glass, Bloomberg, Bernstein analysis

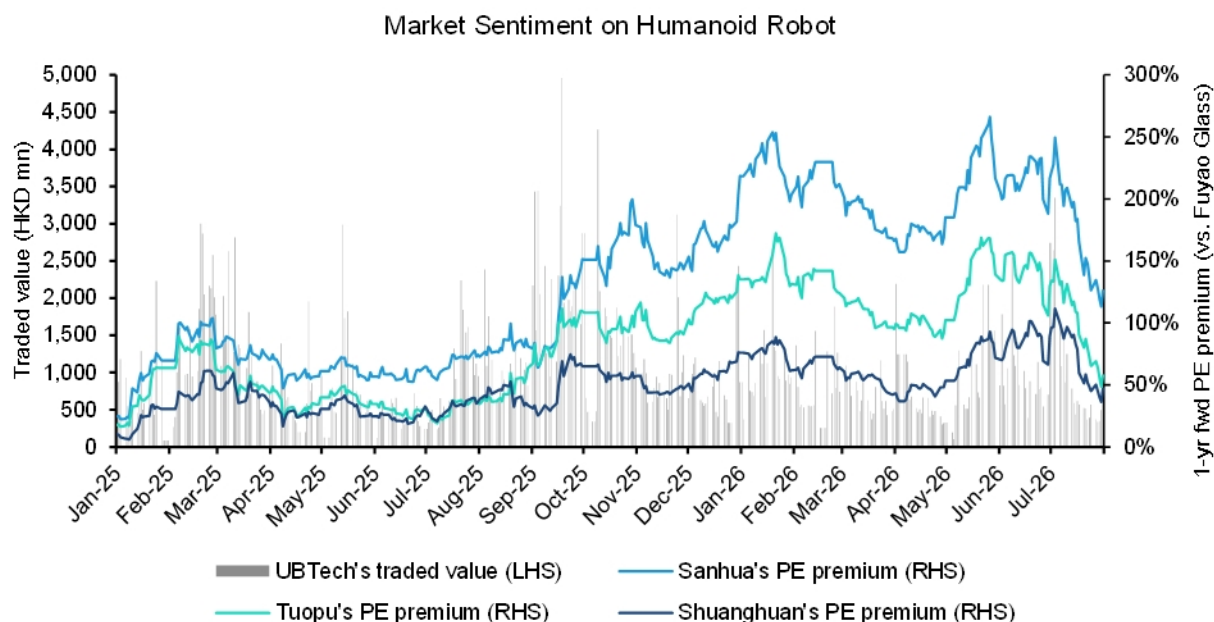
EXHIBIT 3: **Sanhua: Valuation premium attributed to Tesla & humanoid robots**

Sanhua's valuation premium vs. Events of Tesla humanoid robot

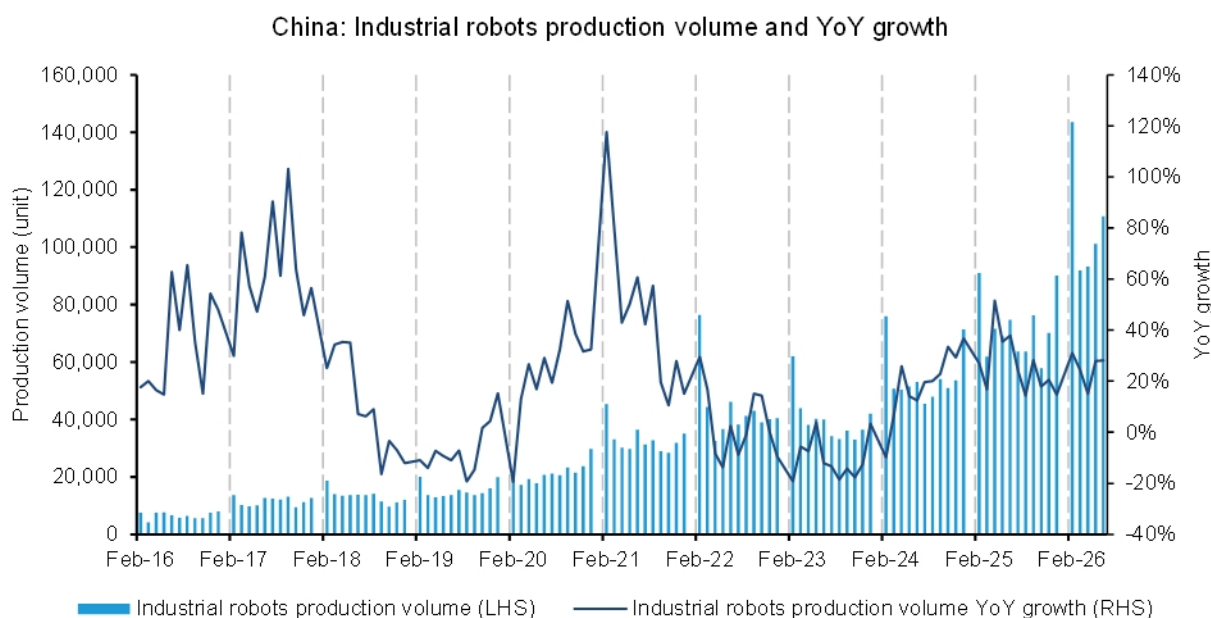


Note: We use Fuyao Glass (600660.CH/3606.HK) as the benchmark. Fuyao Glass is not covered by Bernstein.

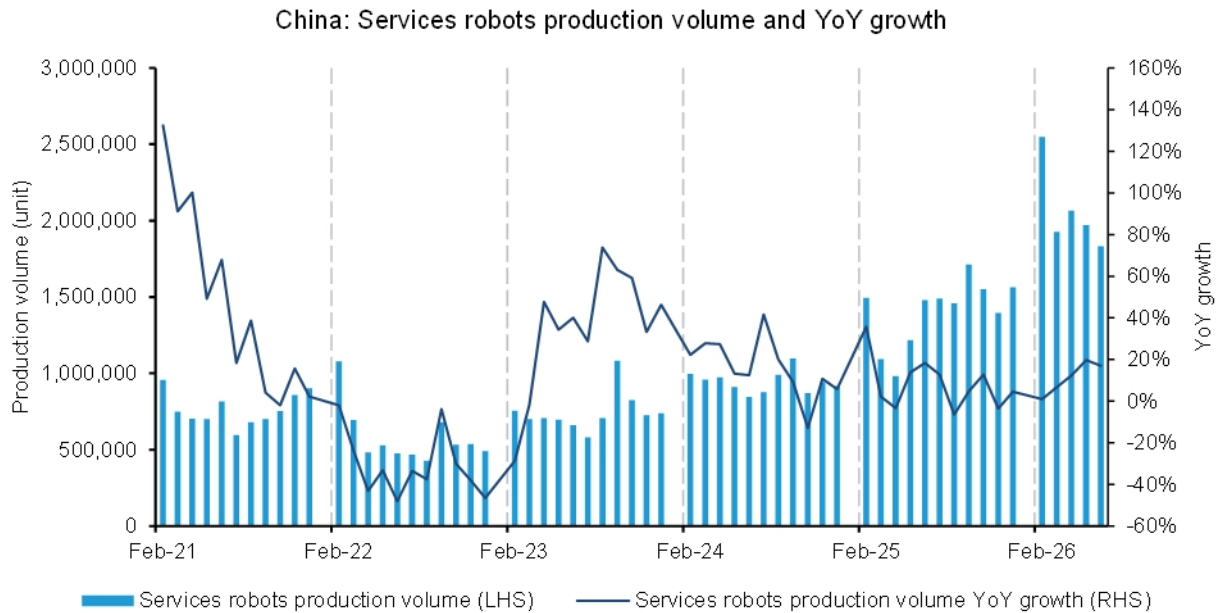
Source: Tesla, industry news, Sanhua, Fuyao Glass, Bloomberg, Bernstein analysis

EXHIBIT 4: **Market sentiment on humanoid robot**

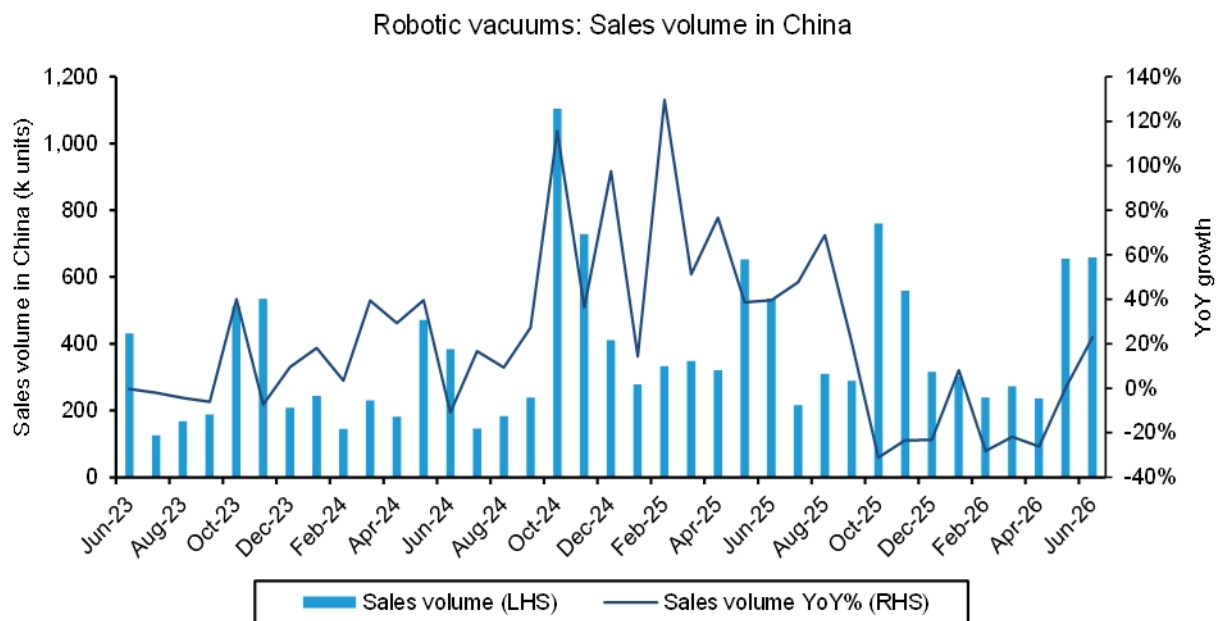
Note: We use Fuyao Glass (600660.CH/3606.HK) as the benchmark. Fuyao Glass and UBTech (9880.HK) are not covered by Bernstein.
 Source: Tesla, industry news, Tuopu, Shuanghuan, Fuyao Glass, UBTech, Bloomberg, Bernstein analysis

EXHIBIT 5: **Production volume of industrial robots in China**

Source: China NBS, Bernstein analysis

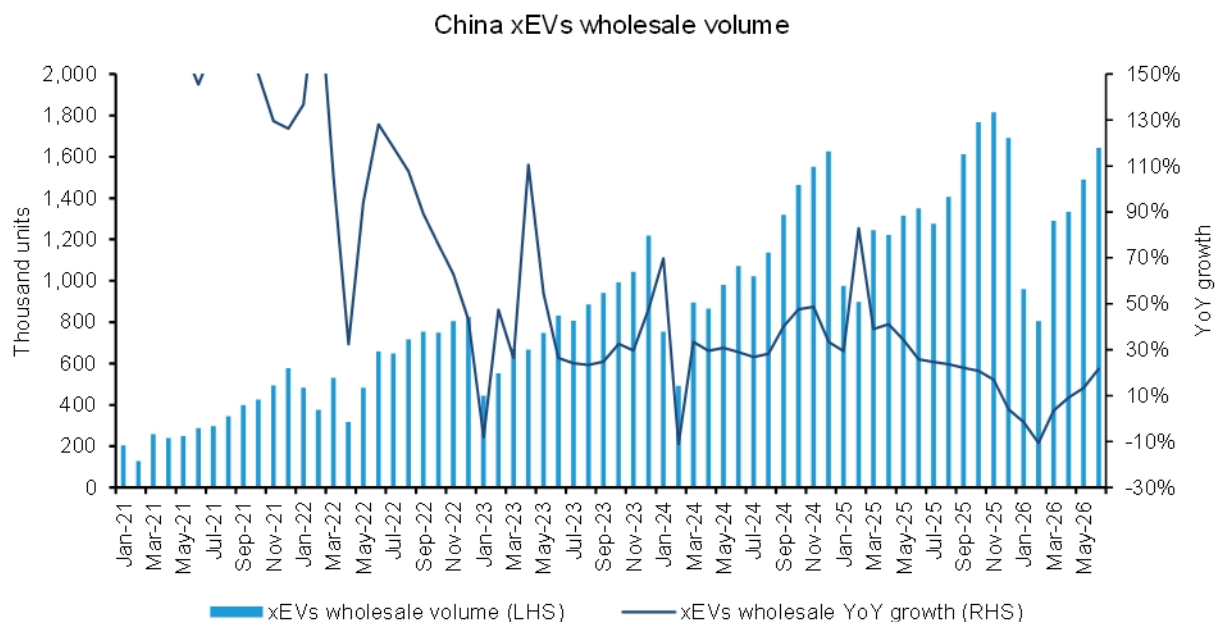
EXHIBIT 6: **Production volume of services robots in China**

Source: China NBS, Bernstein analysis

EXHIBIT 7: **Sales volume of robotic vacuums in China**

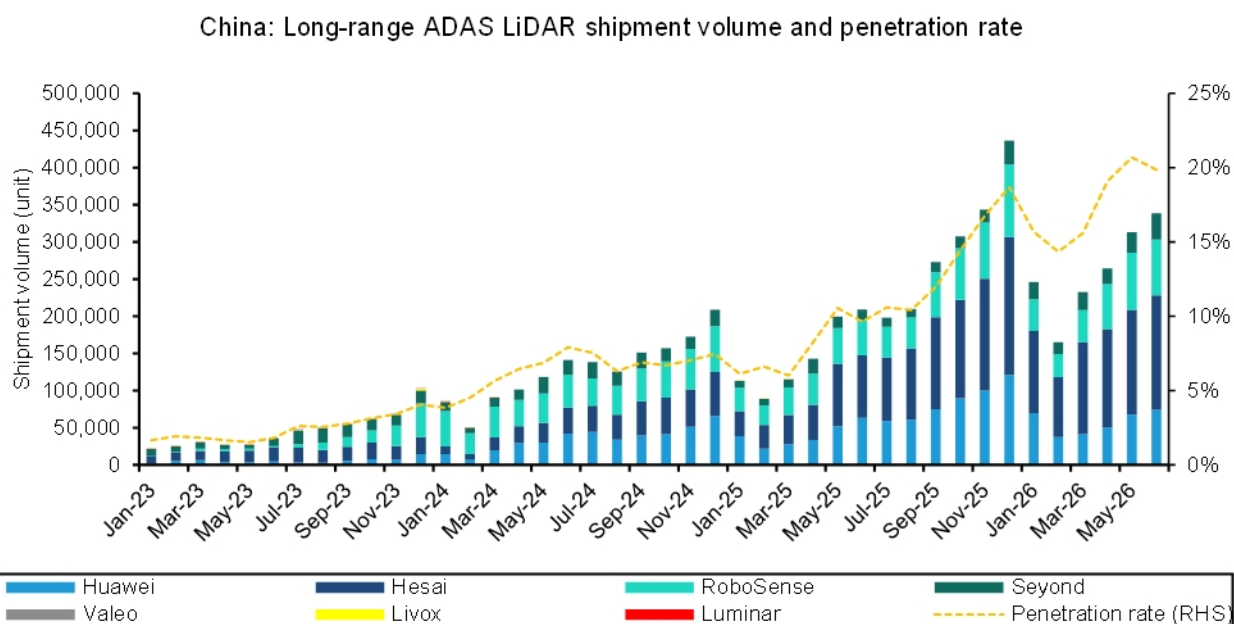
Note: The data covers offline sales channels and online sales channels (excl. Doyin).

Source: AVC, Bernstein analysis

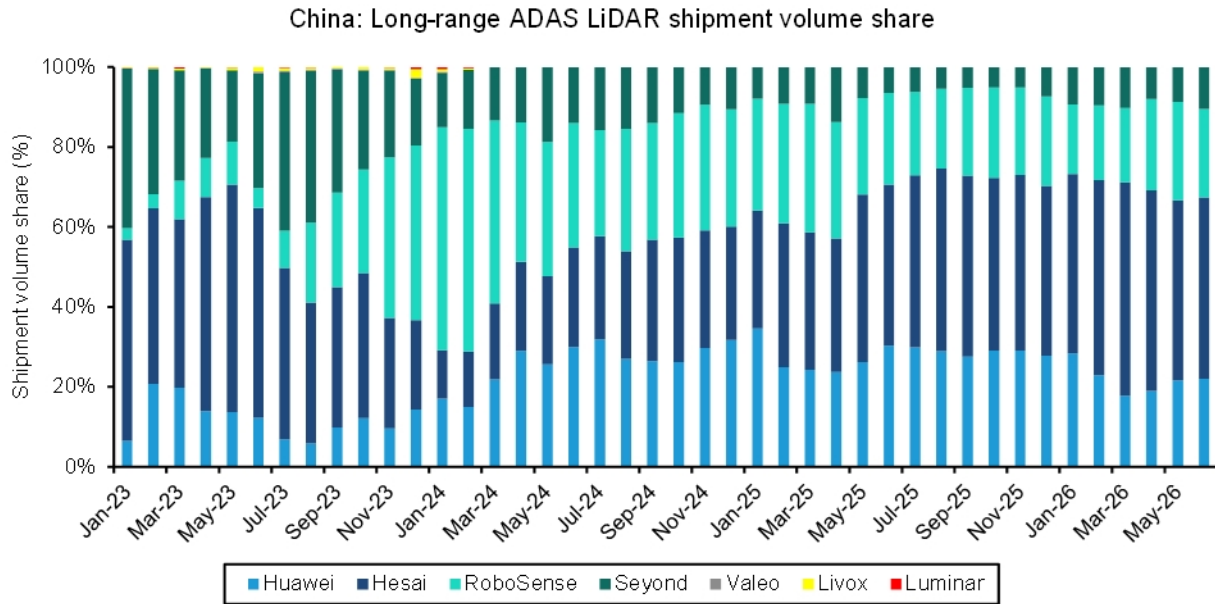
EXHIBIT 8: **China xEVs wholesale volume**

Note: xEVs include BEVs, PHEVs, hybrid and others

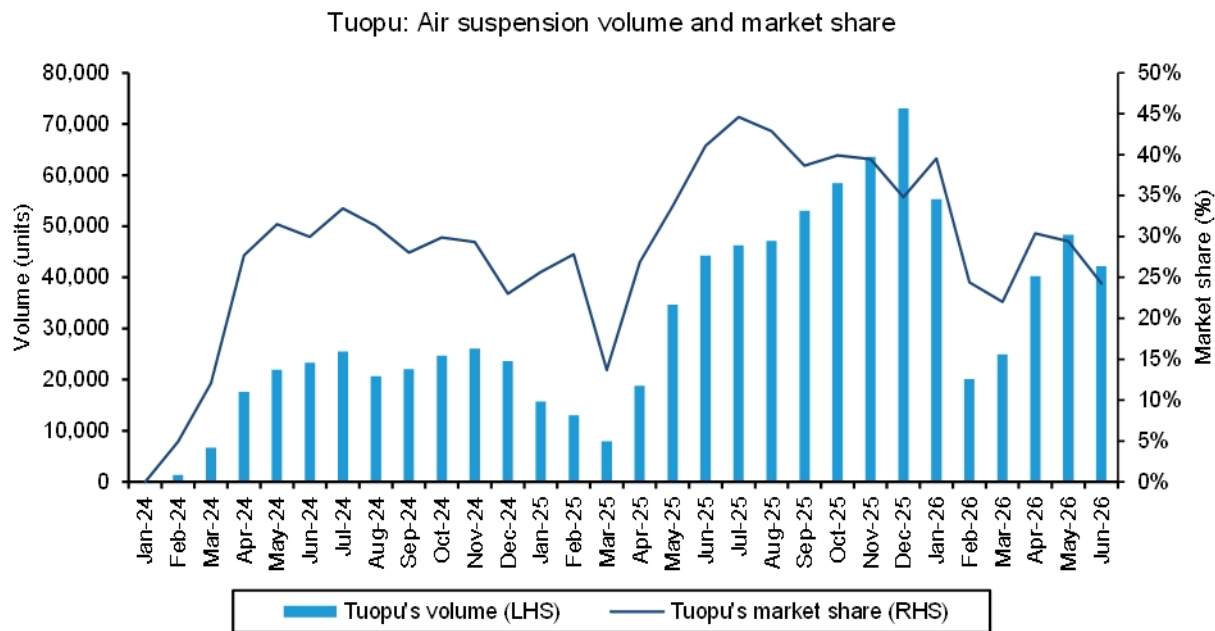
Source: CAAM, Bernstein Asian Autos team, Bernstein analysis

EXHIBIT 9: **Long-range ADAS LiDAR in China: Increasing penetration and volume share breakdown**

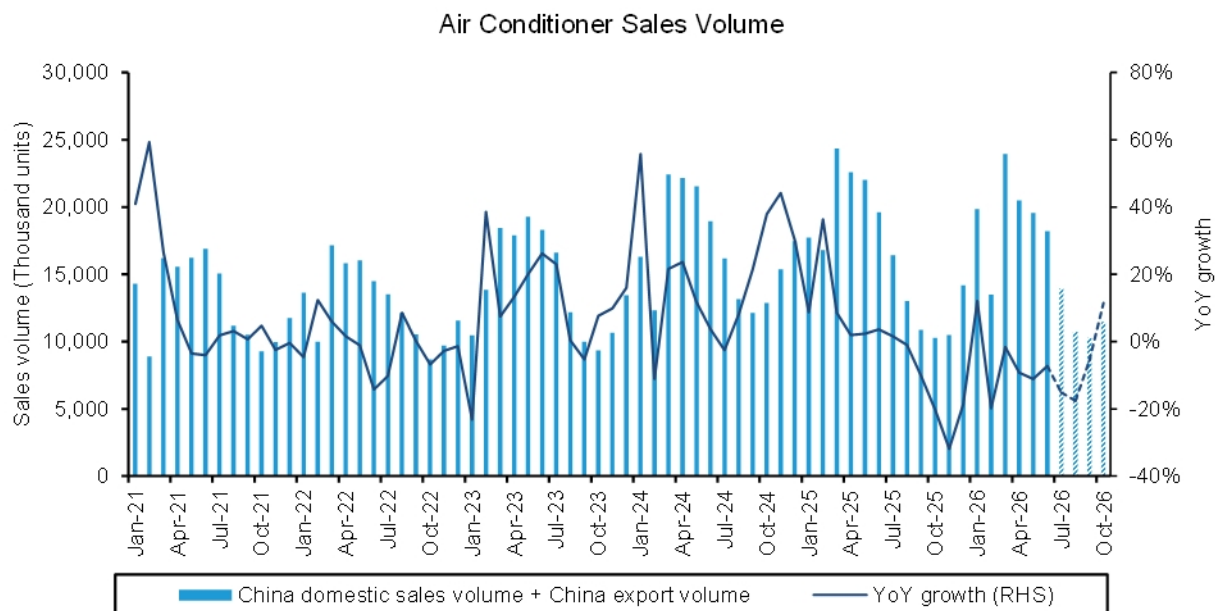
Source: Gasgoo, Bernstein analysis

EXHIBIT 10: **Long-range ADAS LiDAR in China: Market share breakdown**

Source: Gasgoo, Bernstein analysis

EXHIBIT 11: **Tuopu's air suspension volume and market share**

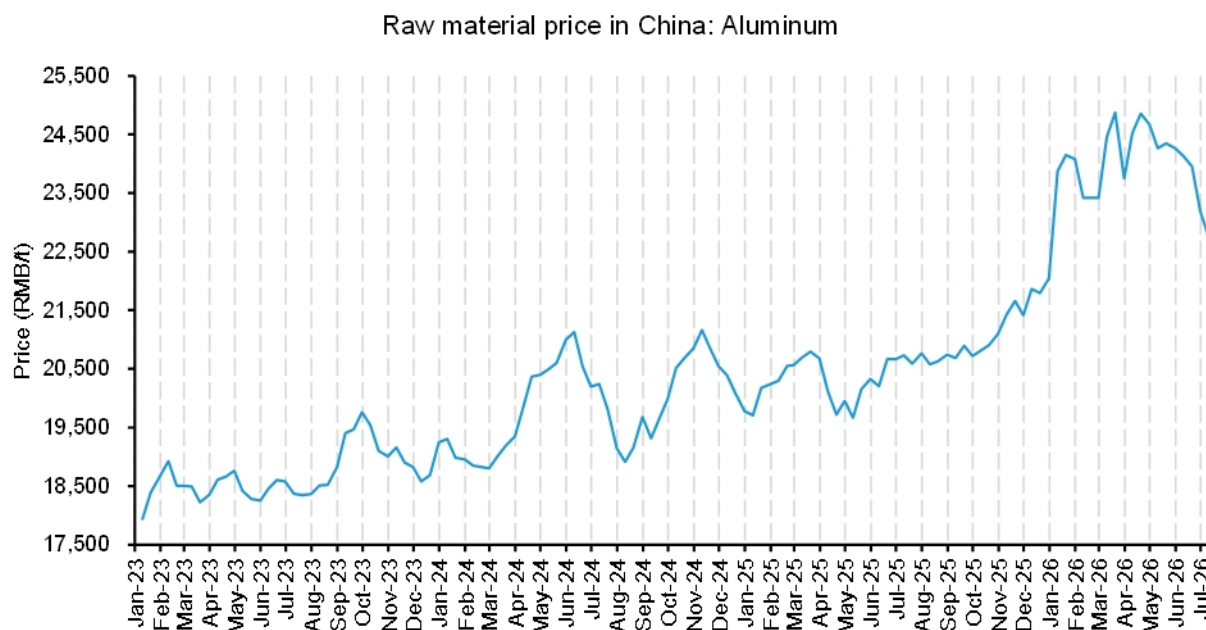
Source: Gasgoo, Bernstein analysis

EXHIBIT 12: **Air conditioner sales volume (China domestic sales + export)**

Source: ChinalOL, Bernstein China Consumer Team, Bernstein analysis

EXHIBIT 13: **Raw material price: Copper**

Source: China NBS, Wind, Bernstein analysis

EXHIBIT 14: **Raw material price: Aluminum**

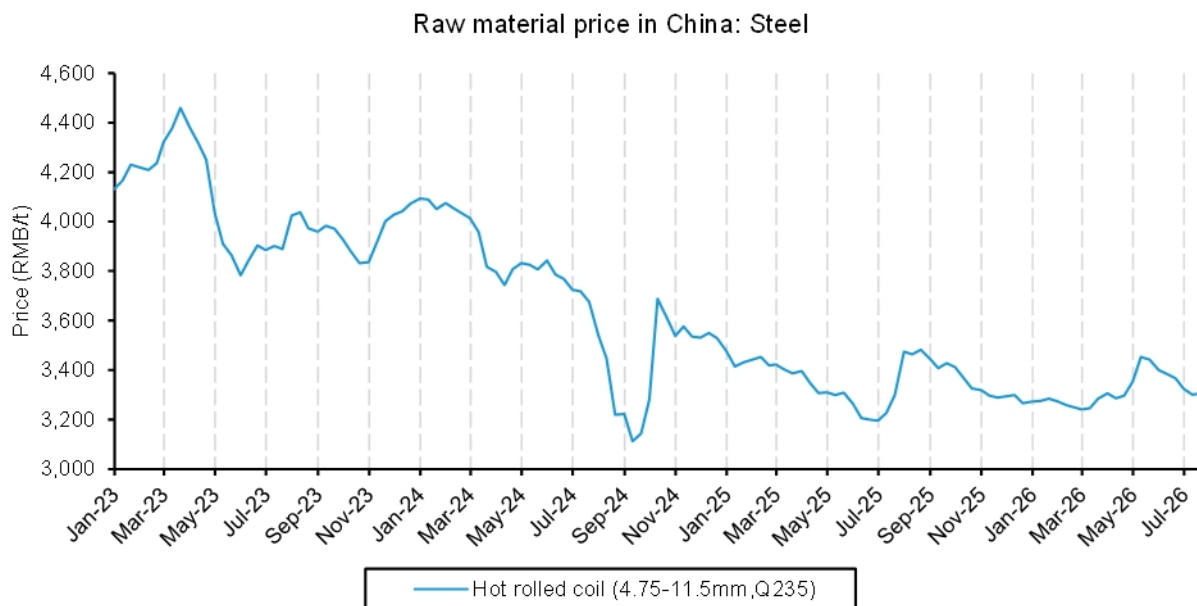
Source: China NBS, Wind, Bernstein analysis

EXHIBIT 15: **Raw material price: Butadiene rubber**

Source: China NBS, Wind, Bernstein analysis

EXHIBIT 16: **Raw material price: Polypropylene**

Source: Wind, Bernstein analysis

EXHIBIT 17: **Raw material price: Steel**

Source: China NBS, Bernstein analysis

EXHIBIT 18: **Raw material price: Gear steel**

Source: Wind, Bernstein analysis

INVESTMENT IMPLICATIONS

We rate Shuanghuan, Hesai and Tuopu Outperform, Sanhua Market-Perform and Leader Drive Underperform.

BERNSTEIN TICKER TABLE

Ticker	Rating	31 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
HSAL.US (Hesai)	O	USD	16.90	30.00	(43.9)%	CNY	2.92	3.31	5.27	39.1	34.5	21.6
2525.HK (Hesai)	O	HKD	16.76	238.00	NA	CNY	2.92	3.31	5.27	4.9	4.4	2.7
688017.CH (Leader Drive)	U	CNY	303.65	115.00	105.3%	CNY	0.69	0.83	0.96	438.5	364.0	315.8
601689.CH (Tuopu)	O	CNY	46.65	75.00	(30.8)%	CNY	1.61	1.81	2.36	29.0	25.8	19.8
2050.HK (Sanhua)	M	HKD	26.18	27.00	(32.7)%	CNY	1.03	0.97	1.11	21.9	23.2	20.4
002050.CH (Sanhua)	M	CNY	37.53	39.00	9.8%	CNY	1.03	0.97	1.11	36.4	38.6	33.9
002472.CH (Shuanghuan)	O	CNY	38.42	60.00	(18.8)%	CNY	1.50	1.70	2.00	25.6	22.6	19.2
ASIA			1,914.70									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

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VALUATION METHODOLOGY**Hesai Group**

We use DCF as the primary valuation method. We set target prices at USD 30.0 and HKD 238.0 for its ADR and H-Share, respectively, based on WACC of 8.3% and terminal growth rate of 3.0%, respectively. They reflect our long term forecast of the company's existing core business and rising contribution from robotics. We use PE method as a reference, with our target price implies a 39x forward PE at end-2026.

Leader Harmonious Drive Systems Co., Ltd.

We use DCF as the primary valuation method. We set target prices at CNY 115.0 based on WACC of 8.3% and terminal growth rate of 3.0%, which reflects our long term forecast of the company's existing core business and rising contribution from robotics. We use PE method as a reference, with our target price implies a 120x forward PE at end-2026.

Ningbo Tuopu Group Co., Ltd.

We use DCF as the primary valuation method. We set target prices at CNY 75.0 based on WACC of 8.0% and terminal growth rate of 3.0%, which reflects our long term forecast of the company's existing core business and rising contribution from robotics. We use PE method as a reference, with our target price implies a 32x forward PE at end-2026.

Zhejiang Sanhua Intelligent Controls Co., Ltd.

We use DCF as the primary valuation method. We set target prices at CNY 39.0 and HKD 27.0 for its A-Share and H-Share, respectively, based on WACC of 8.2% and terminal growth rate of 3.0%. They reflect our long term forecast of the company's existing core business and rising contribution from robotics. We use PE method as a reference, with our target price implies a 35x forward PE at end-2026.

Zhejiang Shuanghuan Driveline Co., Ltd.

We use DCF as the primary valuation method. We set target prices at CNY 60.0 based on WACC of 7.6% and terminal growth rate of 3.0%, which reflects our long term forecast of the company's existing core business and rising contribution from robotics. We use PE method as a reference, with our target price implies a 30x forward PE at end-2026.

RISKS**Hesai Group**

Downside risks are: 1) serious car accidents caused by LiDAR failures; 2) escalated sanctions from the US government; 3) loss of market share at key clients; and 4) slower-than-expected improvements in profitability.

Leader Harmonious Drive Systems Co., Ltd.

Potential upside risks include: 1) official entry into leading humanoid robot makers; 2) breakthroughs or unexpected policy support in Chinese humanoid robot industry; 3) further market share gains in strain-wave reducers among global industrial robot leaders; and 4) proven commercial success of its roller screws.

Ningbo Tuopu Group Co., Ltd.

Downside risks include 1) Tesla fails to meet investors' expectations in automotive or robots; 2) diminished investor interest in humanoid robots due to absence of breakthroughs; 3) a slowdown in vehicle sales; and 4) negative impacts from geopolitical issues (e.g. tariffs, export control of magnetic materials).

Zhejiang Sanhua Intelligent Controls Co., Ltd.

Potential upside risks include: 1) Tesla shows material progress in automotive or robotics; 2) significant progress in data centers and energy storage; and 3) release of specialized heat management solutions for robotics. Downside risks include: 1) Tesla fails to meet investors' expectations in automotives or robotics; 2) diminished investor interest in humanoid robots due to absence of breakthroughs; 3) a slowdown in sales of new energy vehicles and air conditioners; and 4) negative impacts from geopolitical issues (e.g. tariffs, export control of magnetic materials).

Zhejiang Shuanghuan Driveline Co., Ltd.

Downside risks include: 1) Tesla robot fails to meet investors' expectations; 2) diminished investor interest in humanoid robots due to absence of major breakthroughs; 3) slowdown sales of new energy vehicles and robotic vacuums; 4) delays in expansion in Europe or upgrading EV gears from parallel-shaft to coaxial layout;

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- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

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The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR

(EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

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Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

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- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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Autonomous credit ratings are based on a 6-month time horizon.

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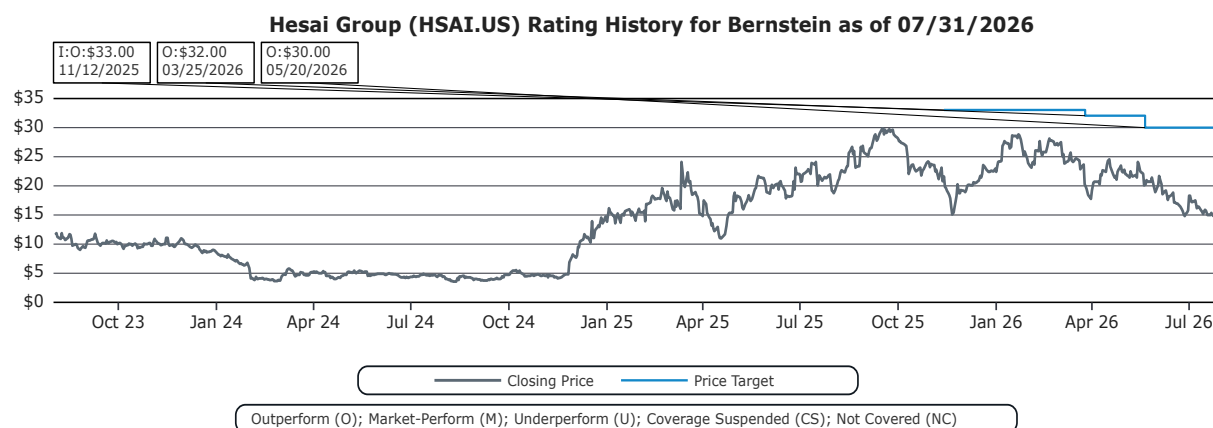
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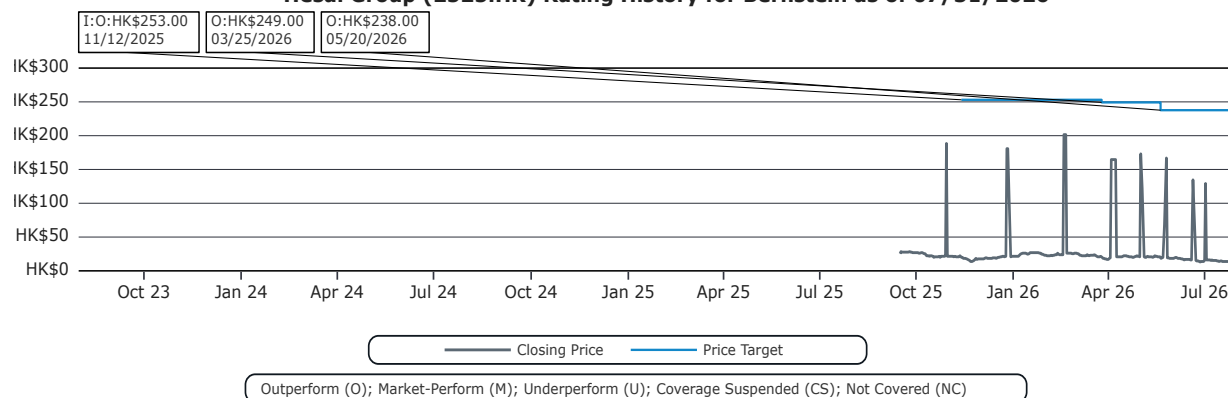
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

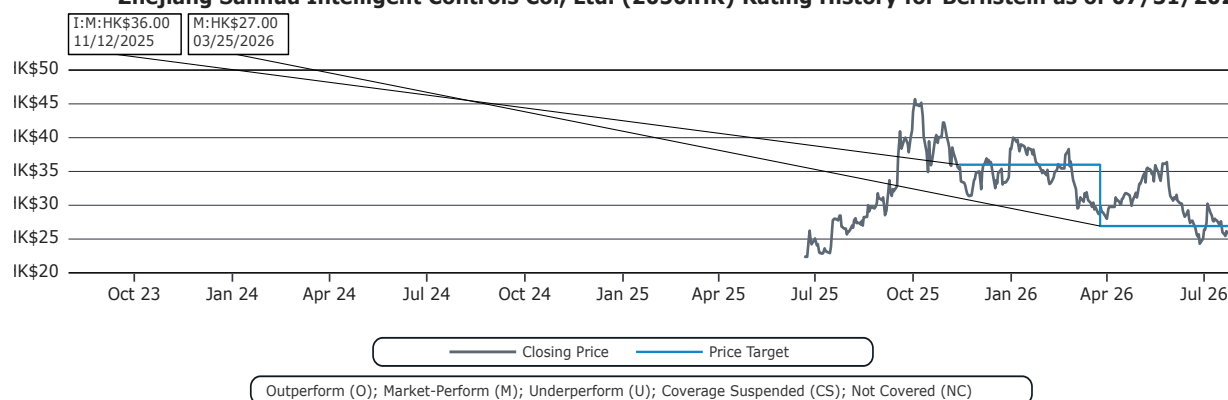
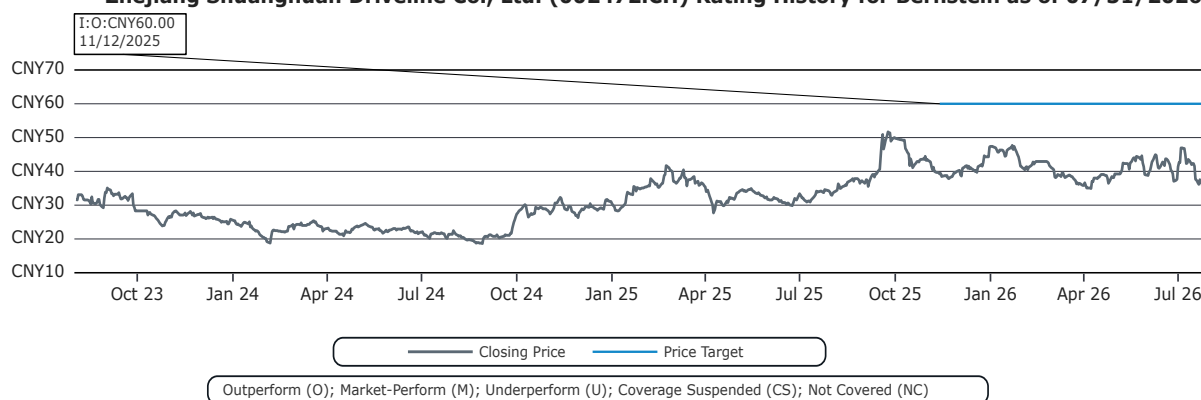
* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



Hesai Group (2525.HK) Rating History for Bernstein as of 07/31/2026**Leader Harmonious Drive Systems Co., Ltd. (688017.CH) Rating History for Bernstein as of 07/31/2026****Ningbo Tuopu Group Co., Ltd. (601689.CH) Rating History for Bernstein as of 07/31/2026**

Zhejiang Sanhua Intelligent Controls Co., Ltd. (2050.HK) Rating History for Bernstein as of 07/31/2026**Zhejiang Sanhua Intelligent Controls Co., Ltd. (002050.CH) Rating History for Bernstein as of 07/31/2026****Zhejiang Shuanghuan Driveline Co., Ltd. (002472.CH) Rating History for Bernstein as of 07/31/2026**

All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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